

AMAN KUMAR SAH – BIOGRAPHY .

My name is **Aman Kumar Sah**, and I am an Artificial Intelligence Engineer and a B.Tech graduate specializing in Artificial Intelligence and Machine Learning. I come from **Birpara, Alipurduar**, located in the famous **Siliguri Corridor of West Bengal**. I am originally from a humble family background and have built myself through consistent practice, discipline, and a strong desire to grow. I have a deep passion for AI, mathematics, and system building, and I aim to contribute significantly to India's technological landscape, particularly in applied AI solutions across healthcare, finance, logistics, and document intelligence.

Physically, I am **5.5 feet (165 cm)** tall, and my weight is around **60 kilograms**. I live a simple lifestyle, enjoy sports, and maintain an active schedule balancing studies, professional commitments, mentorship, and continuous learning. I also enjoy playing the flute, which keeps me grounded, calm, and creatively engaged.

PERSONAL BACKGROUND & EARLY LIFE

I was born and raised in **Birpara**, which is part of the **Alipurduar district** in the northern region of West Bengal. This area also lies near the *Siliguri Corridor*, a narrow passageway connecting mainland India to the North-Eastern states and neighboring countries. Growing up in a relatively remote yet culturally rich area shaped my worldview. It taught me resilience, humility, and the value of hard work.

My family has always been supportive despite being financially humble. I come from a household where education, discipline, and ambition were highly valued, even if resources were limited. My parents ensured I received quality education and encouraged my interest in science and technology from an early age.

I also have a sister who studies at **IISER Bhopal**, which reinforced my academic motivation and pushed me to pursue excellence in AI, mathematics, and research-oriented fields.

ACADEMIC JOURNEY

I completed my schooling at **Sunshine School**, achieving:

- **93% in Class XII (Science)**
- **89% in Class X**

I then pursued my **B.Tech in Computer Science Engineering (Specialization in AIML)** from **Techno Main Salt Lake**, Kolkata, and graduated with a CGPA of **8.1/10**.

Along this journey:

- I developed a strong interest in mathematics, probability, statistics, and linear algebra.
- I got deeply fascinated by machine learning, optimization, neural networks, and the inner mechanics of transformers.
- I eventually started exploring applied AI, LLM pipelines, computer vision solutions, OCR systems, document processing, and agentic automation.

I am also **GATE DA (Data Science and Artificial Intelligence) Qualified**, proving my technical depth in algorithms, mathematical foundations, machine learning theory, linear algebra, and data science concepts.

MENTORSHIP JOURNEY

One of my unique strengths is teaching. Over time, I have mentored **500+ students** in:

- Mathematics (Algebra, Calculus, Linear Algebra, Probability)
- Machine Learning basics
- Neural networks
- Python programming
- Foundations of AI
- Logical reasoning
- Problem-solving

Mentoring gave me a deeper understanding of concepts and strengthened my communication skills. It also helped me build confidence in breaking complex concepts into simple, understandable parts — a skill extremely useful in designing AI agents, documentation, and real-world deployments.

CAREER EXPERIENCE OVERVIEW

My professional experience spans across multiple domains such as healthcare, logistics, insurance, computer vision, document intelligence, and agent-based systems. I have worked at three companies:

1. Kaara Info Systems (2025) – AI Trainee Consultant

Here I designed end-to-end production AI systems integrating:

- FastAPI
- Azure OpenAI GPT-4o Vision
- Document intelligence
- LangGraph multi-agent systems
- Fraud detection engines
- OCR-based data extraction
- Real-time analytics
- PDF generation and automation workflows
- YOLO-based computer vision pipelines

2. Nexus Info – AI Intern (2024)

I built:

- A domain-specific Agentic RAG chatbot
- Vector database architecture using Chroma
- End-to-end document-based Q&A system
- A disease prediction machine learning system
- CSV visualization + RAG hybrid pipeline

3. Arkham Archives Pvt. Ltd – Computer Vision Intern (2023–24)

I worked on:

- 3D image reconstruction
- Computer vision preprocessing
- Model building for game development pipelines
- Collaborative multi-model integration

TECHNICAL SKILLSET

My technical expertise covers a broad spectrum of modern AI technology:

Programming

- Python
- SQL/MySQL
- HTML/CSS (for minor tasks)
- Shell scripting (basic)

AI & Machine Learning

- Regression models
- Classification models
- Clustering, PCA, dimensionality reduction
- Deep learning using PyTorch and TensorFlow
- CNNs, RNNs, Attention Networks
- Vision transformer basics
- Model evaluation and metrics

Generative AI

- LLMs
- Agentic systems
- LangChain, LangGraph, CrewAI
- RAG systems: FAISS, ChromaDB, BM25
- LLM Optimization: LoRA, QLoRA, PEFT
- Prompt engineering
- Document intelligence
- Vision-Language models

Computer Vision

- OpenCV
- MediaPipe (Pose, FaceMesh, Iris Tracking)
- YOLO v5/v8/v11
- Object detection and classification
- Image preprocessing and augmentation

Cloud & Deployment

- AWS
- Azure
- Azure DevOps
- Azure Vision APIs
- Docker
- Deployment pipelines

Backend Engineering

- FastAPI
- Flask
- Django
- REST API creation
- Authentication + middleware
- Integration with AI pipelines
- File processing (PDF, image, CSV)
- SQLite, MySQL

PROJECTS — EXTREMELY DETAILED DESCRIPTIONS

Below are the detailed descriptions of the systems I built across all companies and personal work.

1. Agentic AI-Based Healthcare Insurance Claim Management System

(Kaara Info Systems, 2025)

This was one of the most advanced systems I have built. It handles the entire lifecycle of an insurance claim:

Major Components:

- **FastAPI backend** with modular endpoints
- **OCR Extraction Engine** using Azure GPT-4o Vision
- **Document parser for discharge summaries, medical bills, prescriptions**
- **Adjudicator workflow**
- **Hospital–Patient–TPA chat system**
- **Agentic RAG chatbot** for claim-related Q&A
- **Multi-agent fraud detection engine** using LangGraph
- **PM-JAY dataset integration** for real procedure vs claimed amount comparison

Fraud Detection Engine:

- Reads extracted OCR fields
- Compares with PM-JAY and surgery cost datasets
- Uses a multi-agent LangGraph workflow:
 - Data validation agent
 - Medical code interpretation agent
 - Cost comparison agent
 - Decision summarizer agent

Invoice Generator

- Automatically creates an invoice after Admin approval
- Generates PDF export and email dispatch
- Uses ReportLab + PyMuPDF backend

This system automated complex workflows, reduced fraud, and saved hours of manual verification.

2. Vehicle Damage Detection & Claim Comparison System

(Kaara Info Systems, 2025)

I worked on collecting thousands of training images, preprocessing them, fine-tuning YOLO models, and building a real-time detection system.

Features:

- Image upload + real-time detection
- Classification of damage severity
- Claim amount estimation
- Real-time GPS location verification
- Comparison with claimed data

3. AI-Based Freight Quotation Generation System

(Kaara Info Systems, 2025)

This system automated logistics cost generation.

Pipeline:

- Email ingestion
- Data extraction (dimensions, weight, mode, origin, destination)
- LLM-based validation
- Surcharge rule engine
- Fuel adjustment factor calculation
- Final quote calculation
- PDF export
- Email auto-send

This reduced manual quotation time from hours to minutes.

4. Document Intelligence & Dynamic PDF Auto-Filling System

(Kaara, 2025)

One of my most technically complex systems.

Key Features:

- Azure GPT-4o Vision extraction
- Field coordinate detection using OpenCV
- Structured mapping to PDF templates
- Auto-fill using PyMuPDF and ReportLab
- Support for Aadhaar, PAN, Payslips, Bank Forms
- Multi-language extraction
- Multi-document upload
- Validation agent for cross-field inconsistencies

This saved massive manual labor for enterprises.

5. Agentic RAG System + Vector DB Integration

(Nexus Info, 2024)

I built:

- Chunking
- Vector embedding
- Semantic search
- Contextual reranking
- FastAPI backend
- ChromaDB vector store
- Migration from HuggingFace LLMs to Gemini

It could answer domain-specific questions from documents.

6. AI Baby Monitoring System

(Personal + applied research)

This multi-modal monitoring system used:

- MediaPipe Pose tracking
- FaceMesh for facial structure
- Iris tracking for eye openness & drowsiness
- Librosa audio classification for crying detection

Alerts triggered via SMS and email.

7. 3D Image Generation System for Games

(Arkham Archives)

I worked on:

- 3D reconstruction
 - Geometry mapping
 - Texture mapping
 - Vision pipelines
 - Data preprocessing
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SPORTS ACHIEVEMENTS

- Top 10 in Gymnastics competitions
- Winner of Yoga competition (twice)
- 2nd rank in Badminton (Team of 2)

These activities reflect my discipline, physical awareness, and team spirit.

HOBBIES & PASSIONS

- Playing the flute (meditative and creative pursuit)

- Learning advanced AI mathematics
 - Experimenting with LangGraph and agentic workflows
 - Building utility apps for automation
 - Learning about global economics and stock market patterns
 - Designing clean backend architectures
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PERSONALITY & WORK ETHIC

I am:

- Curious
- Analytical
- Calm under pressure
- Highly disciplined
- Practical, yet imaginative
- Passionate about learning
- Obsessed with understanding concepts deeply

I thrive on:

- Solving hard problems
 - Optimizing systems
 - Improving workflows
 - Teaching and explaining
 - Building for real-world impact
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VISION & FUTURE GOALS

My long-term goal is to:

- Build **AI infrastructure for India**
- Develop **intelligent systems for healthcare, banking, logistics, and governance**

- Create **Agentic AI frameworks** for enterprise-level automation
 - Contribute to **AI safety, transparency, and optimization**
 - Solve complex Indian problems like fraud detection, documentation bottlenecks, inefficient operations, and manual-heavy workflows
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CONTACT INFORMATION

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- Height: **5.5 feet**
- Weight: **60 kg**